Operations Research: Essentials

Every theorem and method of Lectures 1–5, each with a practice question, its solution right underneath and the gotcha; plus the full method for reading everything off a final tableau or a basis B

Operations Research, midterm

2026-11-02. Every number was recomputed exactly (rational basis enumeration, `verify_essentials.py`).

How to use it. Read the purple *Theorem* boxes to know what you may quote and what hypothesis it needs. Read the blue *Method* box, cover the green *Solution*, do the orange *Practice*, compare, then read the red *Gotcha*. Part V is the one you asked for: a dictionary of everything a final tableau or a basis matrix tells you, in both directions.

Conventions of the course (and of this guide). Maximisation. Equational form $\max\{c^\top x : Ax = b,\ x \geq 0\}$ with $\operatorname{rank} A = m$. A basis B is an ordered list of m independent columns; $x_B = B^{-1}b$, $x_N = 0$. The tableau is written *equation style*: basic variables on the left, nonbasic on the right,

$$x_{B(k)} = \beta_k + \sum_{j \in N} q_{kj} x_j, \qquad z = z_0 + \sum_{j \in N} r_j x_j.$$

A reduced cost $r_j > 0$ means "increasing x_j improves z". $y^\top = c_B^\top B^{-1}$ are the multipliers (shadow prices, dual solution).

The running example, used by most practices. A workshop makes two products, $x_1, x_2 \geq 0$ units per day. Resource 1: $2x_1 + x_2 \leq 10$ hours. Resource 2: $x_1 + x_2 \leq 6$ hours. The market takes at most 5 units of product 2: $x_2 \leq 5$. Profits 4 and 3 (hundreds of MAD).

$$\max\ z = 4x_1 + 3x_2 \quad \text{s.t.} \quad 2x_1 + x_2 \leq 10, \quad x_1 + x_2 \leq 6, \quad x_2 \leq 5, \quad x \geq 0.$$

Slacks s_1, s_2, s_3; variable order $(x_1, x_2, s_1, s_2, s_3)$. Answer, for checking: $(x_1, x_2) = (4, 2)$, $z^\star = 22$, $y = (1, 2, 0)$.

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Part I

Station 1: Model

1 Formulate a linear program

Method

(a) **Variables**: one per decision, defined in words with unit and domain ("x_1 = units of product 1 made per day, $x_1 \geq 0$, continuous because divisible").

(b) **Objective**: max or min of a linear expression; state its unit.

(c) **Constraints**: one row per limit, read the coefficients *down the column* of each variable in the data table; state the unit of every row; add the domain constraints.

(d) **Class**: all linear and continuous ⇒ LP; integrality ⇒ IP; a product of variables ⇒ NLP. A percentage requirement ("at least 25 % of the units are product 2") is linear after clearing the denominator: $x_2 \geq 0.25(x_1 + x_2) \iff 0.25x_1 - 0.75x_2 \leq 0$.

Practice

Given. A workshop makes products 1 and 2. Product 1 needs 2 h on machine A and 1 h on machine B; product 2 needs 1 h on A and 1 h on B. Machine A has 10 h per day, machine B 6 h. At most 5 units of product 2 can be sold per day. Profits: 400 and 300 MAD per unit. Everything made is sold; fractional units are fine for planning.

Your task. Formulate, with units, and name the class.

Solution

Variables: $x_1, x_2 \geq 0$ units made per day (continuous). Objective in hundreds of MAD per day: $\max 4x_1 + 3x_2$. Rows: machine A hours $2x_1 + x_2 \leq 10$; machine B hours $x_1 + x_2 \leq 6$; market, units of product 2, $x_2 \leq 5$. LP: linear objective, linear rows, continuous variables. (This is the running example.)

Gotcha

"x_1 = product 1" is not a definition: say *how many units, per what period.* Changing the objective's unit (MAD to hundreds of MAD) changes z only, never the rows. "At most 5 can be sold" is a constraint on x_2, not a reason to add a sales variable, because every unit made is sold.

Part II

Station 2: Represent

2 Canonical and equational forms; substitutions

Theorem: Canonical and equational forms are equivalent

Canonical: $\max\{c^\top x : Ax \leq b,\ x \geq 0\}$. Equational: $\max\{c^\top x : Ax = b,\ x \geq 0\}$. A $\leq$ row gets a slack $s \geq 0$ added; a $\geq$ row gets a surplus subtracted; an $=$ row stays. The original decisions are unchanged and the optimal values coincide (a feasible point of one form gives one of the other

with the same value).

Method

(a) **Domains first.** $t \geq \ell$: set $u = t - \ell \geq 0$. $t \leq 0$: $t = -u$. t free: $t = u - v$, $u, v \geq 0$. Substitute everywhere, including the objective; **keep the constant** that appears.

(b) **Rows.** $a^\top x \leq \beta$: $a^\top x + s = \beta$. $a^\top x \geq \beta$: $a^\top x - s = \beta$. Minimisation: maximise $-f$ and negate the value at the end.

(c) **Order the variables** and write A, b, c in that order. Say the order.

Practice

Given. min $3x - 2y$ s.t. $x + y \geq 4$, $2x - y \leq 6$, $x \geq 1$, $y \in \mathbb{R}$.

Your task. Write an equivalent equational maximisation with nonnegative variables, give the variable order, A, b, c, and say how to recover the original optimal value.

Solution

$u = x - 1 \geq 0$, $y = v - w$ with $v, w \geq 0$. Objective: $\min 3(u+1) - 2(v-w) = \min 3u - 2v + 2w + 3$, so maximise $z = -3u + 2v - 2w - 3$. Rows: $(u+1) + (v-w) - s_1 = 4 \Rightarrow u + v - w - s_1 = 3$; $2(u+1) - (v-w) + s_2 = 6 \Rightarrow 2u - v + w + s_2 = 4$. Order (u, v, w, s_1, s_2):

$$A = \begin{bmatrix} 1 & 1 & -1 & -1 & 0 \\ 2 & -1 & 1 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad c = (-3, 2, -2, 0, 0)^\top, \quad \text{constant } -3.$$

Original minimum $= -(z^\star)$ where $z^\star$ is the maximum of $c^\top x$ *plus* the constant: min $= -(z^\star_{\text{no const}} - 3)$. Say it in one line.

Gotcha

The surplus is *subtracted*. The constant from a substitution is dropped by half the class: the optimal *value* is off by it even though the optimal *point* is right. Multiplying a row by -1 to get $b \geq 0$ is allowed, but it flips the sign of that row's slack column in A: say that you did it.

3 Redundancy, consistency and rank

Theorem: Redundancy test; rank

An inequality is redundant if a **nonnegative combination** of the other rows (plus $x \geq 0$) implies it; removing it leaves the feasible set unchanged. For an equality system, elimination on $[A \mid b]$: a row $0 = 0$ can be dropped, a row $0 = \beta \neq 0$ means infeasible, the number of pivots is rank A. Slack columns form I_m, so after adding slacks rank $A = m$ automatically.

Method

To **prove redundancy**: exhibit the weights $\lambda_i \geq 0$ with $\sum \lambda_i a_i \leq a$ componentwise and $\sum \lambda_i b_i \leq \beta$ (then use $x \geq 0$ for the dominated coefficients). To **disprove** it: exhibit a point satisfying every other row but violating this one.

Practice

Given. The running example plus a fourth row $3x_1 + 2x_2 \leq 17$.

Your task. (a) Is the new row redundant? Prove it. (b) Is $x_2 \leq 5$ redundant? Prove your

answer.

Solution

(a) Yes: $1 \cdot (2x_1 + x_2 \le 10) + 1 \cdot (x_1 + x_2 \le 6)$ gives $3x_1 + 2x_2 \le 16 \le 17$. (b) No: $(0, 6)$ satisfies $2(0)+6 \le 10$, $0+6 \le 6$, $x \ge 0$ but violates $x_2 \le 5$. So the feasible sets differ, and the row cannot be dropped.

Gotcha

"It looks weaker" is not a proof. Weights must be ≥ 0 (a negative weight flips the inequality). Redundancy of an inequality never changes the optimal value, so "removing a redundant constraint changes $z^\star$" is false.

Part III

Station 2b: Bases and basic feasible solutions

4 Basis, BFS, enumeration, certificate

Theorem: Basis and BFS; fundamental theorem

With $\operatorname{rank} A = m$, a **basis** is an ordered choice of m linearly independent columns $B = [A_{j_1} \cdots A_{j_m}]$. Its basic solution: $x_N = 0$, $x_B = B^{-1}b$. It is a **BFS** iff $B^{-1}b \ge 0$; **nondegenerate** iff all m basic values are > 0. A BFS has *at most* m nonzero coordinates (basic ones may be 0).

Fundamental theorem. If the equational LP is feasible and the objective is bounded above, an optimal BFS exists. Hence enumerating the $\le \binom{n}{m}$ bases and comparing the feasible ones is a correct (slow) algorithm.

Used for. justifying enumeration; "a BFS has exactly m nonzeros" is **false** (degeneracy); several bases can give the same point.

Proof idea. From any optimal solution, remove a positive coordinate by moving along a direction in the kernel of the positive columns until a coordinate hits 0; the objective cannot decrease (otherwise unbounded in the other direction); induction on the support.

Method

(a) Rank: spot I_m in the slack columns, or eliminate.

(b) For each column choice: **independence** first (two columns proportional? $\det = 0$?). If dependent, it is not a basis at all.

(c) Set the other variables to 0, solve the $m \times m$ system by substitution, write the *full* vector.

(d) Feasible iff all coordinates ≥ 0. Compute z only for the feasible ones. Best feasible = optimum (fundamental theorem).

(e) **Certificate**: a nonnegative combination of the rows that bounds $c^\top x$ by $z^\star$ (that is the dual solution, Station 4).

Practice

Given. $\max 2x_1 + 3x_2$ s.t. $x_1 + 2x_2 + s_1 = 7$, $3x_1 + x_2 + s_2 = 6$, all variables ≥ 0. Order (x_1, x_2, s_1, s_2).

Your task. (a) rank A. (b) Table of the six column pairs: basis or not, full basic solution, feasible, z. (c) Optimal value, proved by a combination of the rows. (d) Unique?

Solution

(a) Columns 3, 4 are I_2, so $\operatorname{rank} A = 2$. (b) No pair is proportional, so all six are bases.

basic	x_1	x_2	s_1	s_2	feasible, z
x_1, x_2	1	3	0	0	yes, 11
x_1, s_1	2	0	5	0	yes, 4
x_1, s_2	7	0	0	-15	no
x_2, s_1	0	6	-5	0	no
x_2, s_2	0	$\frac{7}{2}$	0	$\frac{5}{2}$	yes, $\frac{21}{2}$
s_1, s_2	0	0	7	6	yes, 0

(x_1, x_2: $x_1 + 2x_2 = 7$, $3x_1 + x_2 = 6 \Rightarrow 5x_1 = 5$.) (c) $z^\star = 11$ at $(1, 3, 0, 0)$. Certificate: $\frac{7}{5}\,\text{row}_1 + \frac{1}{5}\,\text{row}_2$: $(\frac{7}{5}+\frac{3}{5})x_1 + (\frac{14}{5}+\frac{1}{5})x_2 + \frac{7}{5}s_1 + \frac{1}{5}s_2 = \frac{49}{5}+\frac{6}{5} = 11$, i.e. $2x_1 + 3x_2 = 11 - \frac{7}{5}s_1 - \frac{1}{5}s_2 \le 11$ for every feasible point. (d) Unique: $z = 11$ forces $s_1 = s_2 = 0$ (strictly negative coefficients), and then the rows give $(1, 3)$.

Gotcha

A dependent pair is "not a basis", not "infeasible". "The best of the feasible ones" proves optimality only through the fundamental theorem; if the question says *prove*, give the row combination. Where do the weights $\frac{7}{5}, \frac{1}{5}$ come from? They are $y^\top = c_B^\top B^{-1}$ of the optimal basis (Part V).

Part IV

Station 3: Simplex

5 The theorems behind one simplex step

Theorem: Tableau of a basis

For a basis B, define $r^\top = c^\top - c_B^\top B^{-1} A$. For *every* x with $Ax = b$ (feasible or not):

$$x_B = B^{-1}b - B^{-1}A_N x_N, \qquad c^\top x = c_B^\top B^{-1} b + \sum_{j \in N} r_j x_j, \qquad r_B = 0.$$

The tableau is unique: it depends on B only, not on the pivots that reached it.

Used for. everything in Part V; the objective row as a proof.

Proof idea. $Ax = Bx_B + A_N x_N$, solve for x_B, substitute into $c^\top x$.

Theorem: Basic direction and ratio test

Raise nonbasic x_j by $\theta \ge 0$, other nonbasics fixed at 0: direction $d_j = 1$, $d_B = -B^{-1}A_j$ (the displayed column of x_j), $Ad = 0$, gain $c^\top d = r_j$ per unit. Feasibility $x_B + \theta d_B \ge 0$ allows

$$\theta_{\max} = \min_{k:\ d_{B(k)} < 0} \frac{x_{B(k)}}{-d_{B(k)}}.$$

If no $d_{B(k)} < 0$ and $r_j > 0$: **unbounded**. If $\theta_{\max} = 0$: **degenerate** step (basis changes, point does not). Otherwise the pivot exchanges x_j with a row attaining the minimum and the new point is a BFS with z increased by $\theta_{\max} r_j$.

Theorem: Optimality, uniqueness, termination

(1) If the basic solution is feasible and all nonbasic $r_j \le 0$, it is optimal (every feasible x has $c^\top x = z_0 + \sum r_j x_j \le z_0$).

(2) Converse only under nondegeneracy: an optimal *nondegenerate* BFS has all $r_j \le 0$. At a degenerate optimum a basis may show $r_j > 0$ with $\theta_{\max} = 0$.

(3) If all nonbasic $r_j < 0$ strictly, the optimum is unique. If the BFS is nondegenerate and some nonbasic $r_j = 0$, there are infinitely many optima (the segment $0 \le t \le \theta_{\max}$ along that direction).

(4) Without degeneracy every pivot strictly increases z, no basis repeats, so simplex terminates.

6 Run the simplex method by hand

Method

(a) Start from the slack basis if $b \ge 0$: $s = b - Ax$, $z = 0 + c^\top x$.

(b) **Enter**: a variable with $r_j > 0$ (largest, unless told otherwise). None $\Rightarrow$ optimal, stop.

(c) **Ratio test**: only rows where the entering variable has a *negative* displayed coefficient; $\theta = \min(\text{constant}/|\text{coefficient}|)$. No such row $\Rightarrow$ unbounded.

(d) **Pivot**: solve the winning row for the entering variable, substitute into every other row and into z. Write the full new tableau and the BFS.

(e) At the end read $x^\star$ (all variables, slacks included), $z^\star$, uniqueness from the signs, y from the slack reduced costs.

Practice

Given. The running example.

Your task. Solve from the slack basis, entering the largest reduced cost. Show every tableau, ratio test, the BFS after each pivot; then give the full solution, the value, uniqueness, and the dual solution.

Solution

Tableau 0 $s_1 = 10 - 2x_1 - x_2$, $s_2 = 6 - x_1 - x_2$, $s_3 = 5 - x_2$, $z = 0 + 4x_1 + 3x_2$. BFS $(0, 0, 10, 6, 5)$.

Pivot 1 x_1 enters ($r = 4$). Ratios: s_1: $10/2 = 5$; s_2: $6/1 = 6$; s_3: x_1 absent, no limit. s_1 leaves at $\theta = 5$. Solve $x_1 = 5 - \frac{1}{2}x_2 - \frac{1}{2}s_1$ and substitute:

$$x_1 = 5 - \tfrac{1}{2}x_2 - \tfrac{1}{2}s_1, \quad s_2 = 1 - \tfrac{1}{2}x_2 + \tfrac{1}{2}s_1, \quad s_3 = 5 - x_2, \quad z = 20 + x_2 - 2s_1.$$

BFS $(5, 0, 0, 1, 5)$, $z = 20$ (gain $5 \cdot 4$).

Pivot 2 x_2 enters ($r = 1$). Ratios: x_1: $5/\frac{1}{2} = 10$; s_2: $1/\frac{1}{2} = 2$; s_3: $5/1 = 5$. s_2 leaves at $\theta = 2$. Solve $x_2 = 2 + s_1 - 2s_2$:

$$x_1 = 4 - s_1 + s_2, \quad x_2 = 2 + s_1 - 2s_2, \quad s_3 = 3 - s_1 + 2s_2, \quad z = 22 - s_1 - 2s_2.$$

Read $x^\star = (x_1, x_2, s_1, s_2, s_3) = (4, 2, 0, 0, 3)$, $z^\star = 22$. Both reduced costs strictly negative $\Rightarrow$ unique. $y = (1, 2, 0)$: minus the reduced costs of s_1, s_2, and 0 for s_3 because s_3 is basic. The last row is the proof: $z = 22 - s_1 - 2s_2 \le 22$ for every feasible plan.

Gotcha

A row where the entering variable has a *positive* coefficient never limits the step (that variable grows). Forgetting the basic slack in the answer ($s_3 = 3$: resource 3 is not binding) loses the point. A zero reduced cost in an intermediate tableau says nothing about optimality yet.

7 Recognise the special cases in a tableau

Practice

Given. Four tableaux (max convention):
(i) $x_1 = 2 + s_1$, $x_2 = 3 + 2s_1$, $z = 7 + s_1$. (ii) $x_1 = 0 - s_1$, $x_2 = 4 + s_1$, $z = 8 + 2s_1$.
(iii) $x_1 = 4 - \frac{1}{2}x_2 - \frac{1}{2}s_2$, $x_3 = 2 - \frac{1}{2}x_2 - s_1 + \frac{1}{2}s_2$, $z = 14 + 0x_2 - s_1 - s_2$. (iv) $x_1 = -1 + s_1$, $x_2 = 3 - s_1$, $z = 10 - s_1$.

Your task. For each: optimal? unbounded? degenerate? unique? If optima are not unique, list them all.

Solution

(i) $r_{s_1} = 1 > 0$ and no row has a negative coefficient of s_1: **unbounded**, along $(x_1, x_2, s_1) = (2, 3, 0) + t(1, 2, 1)$.
(ii) $r_{s_1} = 2 > 0$ but the x_1 row gives $\theta_{\max} = 0/1 = 0$: **degenerate** BFS $(0, 4, 0)$; the pivot changes the basis, not the point. Not yet known to be optimal; pivot.
(iii) All $r \le 0$ and constants ≥ 0: **optimal**, $z^\star = 14$, BFS $(4, 0, 2, 0, 0)$ nondegenerate, $r_{x_2} = 0$: **infinitely many optima**. Set $x_2 = t$, $s_1 = s_2 = 0$: $x_1 = 4 - \frac{t}{2} \ge 0$, $x_3 = 2 - \frac{t}{2} \ge 0 \Rightarrow 0 \le t \le 4$. Optima: $(4 - \frac{t}{2},\ t,\ 2 - \frac{t}{2})$, from $(4, 0, 2)$ to $(2, 4, 0)$, all with $z = 14$.
(iv) Objective row fine but $x_1 = -1 < 0$: **infeasible basis**, not optimal. (A dual simplex pivot would repair it, Part VII.)

Gotcha

Check *both* conditions: constants ≥ 0 and reduced costs ≤ 0. "No limiting row" (unbounded) and "limit 0" (degenerate) are different. A zero reduced cost at a *degenerate* optimum does not by itself give a segment of optima: $\theta_{\max}$ may be 0.

Part V

Reading everything from a final tableau or a basis B

8 The dictionary: tableau $\leftrightarrow$ matrices

Method: the two directions

Direction 1, you have B (a list of basic variables in order). Compute, in this order:

1. B = those columns of A (original A with slack columns), in the stated order. c_B = their costs.
2. B^{-1} (solve $Bq = v$ or use the 2×2 formula $\frac{1}{\det}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$).
3. $x_B = B^{-1}b$ (constants). Feasible iff ≥ 0. $z_0 = c_B^\top x_B$.

4. $y^\top = c_B^\top B^{-1}$ (multipliers).
5. For each nonbasic j: $r_j = c_j - y^\top A_j$; displayed column $= -B^{-1}A_j$.
6. Optimal iff $x_B \geq 0$ and all $r_j \leq 0$.

Direction 2, you have the final tableau. Read, without computing:

you want	where it is in the equation-style tableau
basis, order	the left-hand sides, top to bottom
$x^\star$	basic = constants; every nonbasic = 0 (slacks included)
$z^\star$	the constant of the z row
binding constraints	those whose slack is nonbasic (or basic at value 0)
reduced cost r_j	coefficient of nonbasic x_j in the z row (basic: 0)
$B^{-1}A_j$	**minus** the displayed column of x_j in the basic rows
B^{-1}	**minus** the slack columns $s_1, \ldots, s_m$ side by side; a *basic* slack s_i contributes the unit column e_k where k is its row position
y_i (shadow price, dual value)	**minus** the reduced cost of s_i; 0 if s_i is basic
dual slack $t_j = (A^\top y)_j - c_j$	$-r_j$ for an original variable x_j (how far its profit is from being worth making)
basic direction of x_j	$d_B =$ its displayed column, $d_j = 1$; gain r_j per unit
$\theta_{\max}$ for x_j	ratio test over the rows with a negative coefficient of x_j
uniqueness	all nonbasic $r_j < 0$: unique; some $r_j = 0$ at a nondegenerate BFS: segment
degeneracy	a basic constant equal to 0
unboundedness	$r_j > 0$ with no negative coefficient in its column

Why the minus signs. The theorem says $x_B = B^{-1}b - B^{-1}A_N x_N$; the tableau puts the nonbasics on the right with their sign, so what you *see* is $-B^{-1}A_j$. A slack s_i has column e_i and cost 0, so its column shows $-B^{-1}e_i$ (the ith column of $-B^{-1}$) and its reduced cost is $0 - c_B^\top B^{-1}e_i = -y_i$.

Gotcha

Other books lay the tableau out differently. In the "z-row" layout (rows $x_B + B^{-1}A_N x_N = B^{-1}b$, bottom row $z + \sum \bar{c}_j x_j = z_0$ with $\bar{c}_j = y^\top A_j - c_j$), the slack columns show $+B^{-1}$, the bottom row shows $+y$ under the slacks, and "$\bar{c}_j \geq 0$ everywhere" is the optimality test. Same numbers, opposite signs. Before reading any tableau, write which layout it is. In a minimisation the improving sign flips again.

Practice

Given. The final tableau of $\max 3x_1 + 2x_2 + x_3$ s.t. $x_1 + x_2 + x_3 \leq 6$, $2x_1 + x_2 \leq 8$, $x \geq 0$ (slacks s_1, s_2):

$$x_1 = 4 - \tfrac{1}{2}x_2 - \tfrac{1}{2}s_2, \qquad x_3 = 2 - \tfrac{1}{2}x_2 - s_1 + \tfrac{1}{2}s_2, \qquad z = 14 + 0\,x_2 - s_1 - s_2.$$

Your task. Read off: (a) the ordered basis and the full $x^\star$; (b) $z^\star$ and which constraints bind; (c) B, B^{-1}, and check $BB^{-1} = I$; (d) y and verify it is dual feasible with $b^\top y = z^\star$; (e) the dual slack of x_2 and what it means; (f) $B^{-1}A_2$ for the column of x_2; (g) is the optimum unique? list all optima; (h) the shadow price and range of resource 1; (i) the range of resource 2; (j) how much the profit of x_3 may change; (k) a new product with column $(1, 1)$ and profit $\frac{5}{2}$: make it or not, and the new optimum if yes.

Solution

(a) Basis (x_1, x_3) in that order; nonbasic x_2, s_1, s_2. $x^\star = (x_1, x_2, x_3, s_1, s_2) = (4, 0, 2, 0, 0)$.

(b) $z^\star = 14$. Both slacks nonbasic (zero): both constraints bind ($4 + 0 + 2 = 6$, $8 + 0 = 8$).

(c) $B = [A_1\ A_3] = \begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}$. Slack columns displayed: $s_1 \to (0, -1)^\top$, $s_2 \to (-\frac{1}{2}, \frac{1}{2})^\top$, so $-B^{-1} = \begin{bmatrix} 0 & -\frac{1}{2} \\ -1 & \frac{1}{2} \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{bmatrix}$. Check: $\begin{bmatrix} 1 & 1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Also $B^{-1}b = (4, 2)^\top = x_B$.

(d) $y = (1, 1)$, minus the slack reduced costs; or $c_B^\top B^{-1} = (3, 1)B^{-1} = (1, 1)$. Dual: $\min 6y_1 + 8y_2$ s.t. $y_1 + 2y_2 \geq 3$, $y_1 + y_2 \geq 2$, $y_1 \geq 1$, $y \geq 0$. Check: $3 \geq 3$, $2 \geq 2$, $1 \geq 1$; $b^\top y = 14 = z^\star$. Optimal pair.

(e) $t_2 = -r_2 = 0$: the dual row of x_2 is tight ($y_1 + y_2 = 2 = c_2$). Product 2 exactly breaks even: making it neither gains nor loses.

(f) $B^{-1}A_2 = -$(displayed column of x_2) $= (\frac{1}{2}, \frac{1}{2})^\top$. Check: $B(\frac{1}{2}, \frac{1}{2})^\top = (1, 1)^\top = A_2$.

(g) Not unique: $r_{x_2} = 0$ at a nondegenerate BFS. $x_2 = t$, $s_1 = s_2 = 0$: $x_1 = 4 - \frac{t}{2}$, $x_3 = 2 - \frac{t}{2}$, so $0 \leq t \leq 4$: optima $(4 - \frac{t}{2},\ t,\ 2 - \frac{t}{2})$, from $(4, 0, 2)$ to $(2, 4, 0)$, value 14.

(h) $y_1 = 1$. $g = B^{-1}e_1 = (0, 1)^\top$ (minus the s_1 column): new constants $x_1 = 4$, $x_3 = 2 + \delta \geq 0 \Rightarrow \delta \geq -2$, no upper limit. Range $b_1 \in [4, \infty)$, $z^\star = 14 + \delta$ there.

(i) $g = B^{-1}e_2 = (\frac{1}{2}, -\frac{1}{2})^\top$: $4 + \frac{\delta}{2} \geq 0$, $2 - \frac{\delta}{2} \geq 0 \Rightarrow -8 \leq \delta \leq 4$, slope $y_2 = 1$.

(j) $c_3 = 1 + \varepsilon$, x_3 basic: add $\varepsilon \times$(x_3 row) to the z row: $z = 14 + 2\varepsilon + (-\frac{\varepsilon}{2})x_2 + (-1 - \varepsilon)s_1 + (-1 + \frac{\varepsilon}{2})s_2$. All ≤ 0: $\varepsilon \geq 0$, $\varepsilon \geq -1$, $\varepsilon \leq 2$. Range $0 \leq \varepsilon \leq 2$: the lower end is 0 because $r_{x_2} = 0$ already; any decrease makes x_2 strictly better.

(k) $r_u = \frac{5}{2} - y^\top a = \frac{5}{2} - 2 = \frac{1}{2} > 0$: make it. Column $-B^{-1}a = -(\frac{1}{2}, \frac{1}{2})^\top$: $x_1 = 4 - \frac{u}{2}$, $x_3 = 2 - \frac{u}{2}$; x_3 leaves at $u = 4$. Pivot: $u = 4 - x_2 - 2x_3 - 2s_1 + s_2$, $x_1 = 2 + x_3 + s_1 - s_2$, $z = 16 - \frac{1}{2}x_2 - x_3 - 2s_1 - \frac{1}{2}s_2$. New optimum $(x_1, x_2, x_3, u) = (2, 0, 0, 4)$, $z^\star = 16$, unique, $y = (2, \frac{1}{2})$.

Gotcha

Rows must be read in the *stated basic order*: B^{-1}'s rows follow the rows of the tableau, its columns follow $s_1, \ldots, s_m$. When a slack is basic its column is not displayed: insert a unit vector. $t_j = -r_j$ only for original variables; for a slack, $-r_{s_i}$ is y_i itself. The range of a basic cost can have 0 as an endpoint (part j): that is not an error, it is a tie.

9 From B to the whole tableau (direction 1, worked)

Practice

Given. The running example, basis (x_1, x_2, s_3) in that order.

Your task. Compute B, B^{-1}, x_B, z_0, y, every reduced cost and the displayed columns, decide whether the basis is optimal, and write the tableau.

Solution

$B = [A_{x_1}\ A_{x_2}\ A_{s_3}] = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $c_B = (4, 3, 0)$. Solve $Bq = v$: $2q_1 + q_2 = v_1$, $q_1 + q_2 = v_2$, $q_2 + q_3 = v_3$ give $q_1 = v_1 - v_2$, $q_2 = 2v_2 - v_1$, $q_3 = v_3 - 2v_2 + v_1$, hence

$$B^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 0 \\ 1 & -2 & 1 \end{bmatrix}, \quad x_B = B^{-1}(10, 6, 5)^\top = (4, 2, 3)^\top \geq 0, \quad z_0 = 4 \cdot 4 + 3 \cdot 2 = 22.$$

$y^\top = c_B^\top B^{-1} = (4 - 3,\ -4 + 6,\ 0) = (1, 2, 0)$. Nonbasic s_1 (column e_1, cost 0): $r = 0 - y_1 = -1$, displayed column $-B^{-1}e_1 = (-1, 1, -1)^\top$. Nonbasic s_2: $r = -y_2 = -2$, column $-B^{-1}e_2 = (1, -2, 2)^\top$. Both $r \leq 0$ and $x_B \geq 0$: optimal. Tableau: $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$, the same as the one simplex reached.

Gotcha

B is built from the *original* A (with slack columns), never from a later tableau. A basic slack (s_3) has $r = 0$ and $y_3 = 0$ by complementary slackness: its resource is not binding ($s_3 = 3$).

Part VI

Station 4: Certify (duality)

10 The theorems

Theorem: Canonical dual; weak and strong duality

Primal $\max\{c^\top x : Ax \le b, x \ge 0\}$, dual $\min\{b^\top y : A^\top y \ge c, y \ge 0\}$. One primal row $\to$ one dual variable; one primal variable $\to$ one dual row.

Weak duality. Primal-feasible x, dual-feasible y: $c^\top x \le y^\top Ax \le b^\top y$. So a feasible pair with $c^\top x = b^\top y$ is optimal on both sides (**matching pair**).

Strong duality. If the primal is feasible with bounded objective, both have optimal solutions and the values are equal. The optimal y is $c_B^\top B^{-1}$ of an optimal basis (read off the final tableau).

Proof idea. Weak: $c \le A^\top y$ and $x \ge 0$ give $c^\top x \le (A^\top y)^\top x = y^\top Ax$; $Ax \le b$ and $y \ge 0$ give $y^\top Ax \le y^\top b$. Strong: the final tableau's $r \le 0$ on original columns says $A^\top y \ge c$, on slack columns says $y \ge 0$, and $b^\top y = c_B^\top B^{-1} b = z^\star$.

Theorem: Dualization rules, max primal / min dual

primal row	dual variable	primal variable	dual row
$a_i^\top x \le b_i$	$y_i \ge 0$	$x_j \ge 0$	$A_j^\top y \ge c_j$
$a_i^\top x \ge b_i$	$y_i \le 0$	$x_j \le 0$	$A_j^\top y \le c_j$
$a_i^\top x = b_i$	$y_i \in \mathbb{R}$	$x_j \in \mathbb{R}$	$A_j^\top y = c_j$

Dual objective $\min b^\top y$. The dual of the dual is the primal. Reason: choose the sign of y_i so that $y_i a_i^\top x \le y_i b_i$ stays true; choose the row direction so that $c_j x_j \le (A_j^\top y) x_j$ for the allowed sign of x_j.

Theorem: Complementary slackness; the outcome table

Primal slack $s = b - Ax$, dual slack $t = A^\top y - c$. For feasible x, y: $b^\top y - c^\top x = \sum_i y_i s_i + \sum_j x_j t_j$, every term ≥ 0. Hence x, y are both optimal iff $y_i s_i = 0\ \forall i$ and $x_j t_j = 0\ \forall j$: a positive dual price needs a tight row; a positive activity needs a tight dual row. Tightness alone forces nothing.
Outcomes. One side finite $\Rightarrow$ both finite and equal. Primal unbounded $\Rightarrow$ dual infeasible. Dual unbounded $\Rightarrow$ primal infeasible. Primal infeasible $\Rightarrow$ dual unbounded *or* infeasible (both can be infeasible).

11 Write a dual with mixed signs

Method

Rule 0: get a *max* primal (negate a min objective). Then one row of the table per primal row, one per primal variable; dual objective $b^\top y$ with the *original* right-hand sides; coefficients of dual row j are read *down* column j of A. Alternative: convert the primal to canonical form first and dualise that; both give equivalent duals.

Practice

Given. $\max 2x_1 - x_2 + 3x_3$ s.t. $x_1 + x_2 + x_3 \le 10$, $x_1 - x_3 \ge 2$, $2x_1 + x_2 = 6$, $x_1 \ge 0$, $x_2 \le 0$, $x_3 \in \mathbb{R}$.

Your task. Write the dual and justify every sign and direction.

Solution

Rows: $\le$ gives $y_1 \ge 0$; $\ge$ gives $y_2 \le 0$; $=$ gives $y_3 \in \mathbb{R}$. Columns: $x_1 \ge 0$: $y_1 + y_2 + 2y_3 \ge 2$; $x_2 \le 0$: $y_1 + 0y_2 + y_3 \le -1$; $x_3 \in \mathbb{R}$: $y_1 - y_2 + 0y_3 = 3$. Dual: $\min 10y_1 + 2y_2 + 6y_3$ subject to those three rows and $y_1 \ge 0$, $y_2 \le 0$, y_3 free.

Gotcha

A $\ge$ row in a max primal gives a *nonpositive* dual variable (multiplying by $y_i \le 0$ turns $\ge$ into $\le$). The dual row of a free variable is an equality; the dual variable of an equality row is free. The dual objective keeps b as written, even $b_2 = 2$ from the $\ge$ row.

12 Certify a candidate with complementary slackness

Method

(a) Check the candidate $\hat{x}$ is primal feasible (every row, every sign). If not, stop: not optimal.

(b) Positive $\hat{x}_j \Rightarrow$ dual row j is an equation. Slack row i ($s_i > 0$) $\Rightarrow y_i = 0$.

(c) Solve the resulting linear system for y.

(d) **Verify** y: signs and the *remaining* dual rows. Then $b^\top y = c^\top \hat{x}$ closes the proof. If the system has no feasible y, $\hat{x}$ is not optimal.

Practice

Given. The running example. Candidates $\hat{x} = (4, 2)$ and $\tilde{x} = (5, 0)$.

Your task. Decide optimality of each by complementary slackness.

Solution

Dual: $\min 10y_1 + 6y_2 + 5y_3$ s.t. $2y_1 + y_2 \geq 4$, $y_1 + y_2 + y_3 \geq 3$, $y \geq 0$.
$\hat{x} = (4, 2)$: feasible ($10 \leq 10$, $6 \leq 6$, $2 \leq 5$). Row 3 slack ($s_3 = 3$) $\Rightarrow y_3 = 0$. $\hat{x}_1, \hat{x}_2 > 0 \Rightarrow 2y_1 + y_2 = 4$, $y_1 + y_2 = 3 \Rightarrow y = (1, 2, 0)$. Verify: $y \geq 0$, both rows tight; $b^\top y = 10 + 12 = 22 = c^\top \hat{x}$. Optimal.
$\tilde{x} = (5, 0)$: feasible ($10 \leq 10$, $5 \leq 6$, $0 \leq 5$). Rows 2, 3 slack $\Rightarrow y_2 = y_3 = 0$. $\tilde{x}_1 > 0 \Rightarrow 2y_1 = 4$, $y_1 = 2$. Verify the dual row of x_2: $y_1 + y_2 + y_3 = 2 \geq 3$ fails. No certificate exists, so $\tilde{x}$ is not optimal (indeed $z = 20 < 22$).

Gotcha

Complementary slackness *finds* the candidate y; only feasibility of both plus equal values *proves* optimality. A tight primal row does not force $y_i > 0$ (both can be zero). Applying the conditions to an infeasible $\hat{x}$ proves nothing.

13 Classify a primal–dual pair

Practice

Given. (a) $\max x_1 + x_2$ s.t. $x_1 - x_2 \leq 1$, $x \geq 0$. (b) $\max x_1$ s.t. $x_1 \leq 1$, $x_1 \geq 2$, $x_1 \geq 0$.

Your task. Classify each primal and its dual, with proof.

Solution

(a) Ray (t, t) is feasible for all $t \geq 0$ ($t - t = 0 \leq 1$) with $z = 2t \to \infty$: **unbounded**. Dual: $\min y$ s.t. $y \geq 1$, $-y \geq 1$, $y \geq 0$: the second row forces $y \leq -1$: **infeasible**, as the theorem predicts. (b) $x_1 \leq 1$ and $x_1 \geq 2$: **infeasible**. Dual (max primal, rows $\leq$ and $\geq$): $\min y_1 + 2y_2$ s.t. $y_1 + y_2 \geq 1$, $y_1 \geq 0$, $y_2 \leq 0$: take $y_1 = 1 + k$, $y_2 = -k$... value $1 + k - 2k = 1 - k \to -\infty$: **unbounded**. The table allows this (infeasible primal, unbounded dual).

Gotcha

"Primal infeasible $\Rightarrow$ dual unbounded" is **false** as a rule (both may be infeasible: $\max x_1$ s.t. $x_1 - x_2 \leq -1$, $-x_1 + x_2 \leq -1$). An unbounded feasible *region* does not mean an unbounded *objective* ($\max -x_1$, $x_1 \geq 0$ has optimum 0).

Part VII

Station 5: Update (sensitivity analysis)

14 The theorems

Theorem: Which test each change can break

Optimality of B = **feasibility** $B^{-1}b \geq 0$ **and optimality** $r^\top = c^\top - c_B^\top B^{-1}A \leq 0$.

change	unchanged	can fail	if it fails
$b \to b'$	B^{-1}, y, all r_j	$B^{-1}b' \geq 0$	dual simplex pivot
$c \to c'$	$B^{-1}b$ (the plan)	$r' \leq 0$	ordinary pivot
new column a, profit γ	everything old	$r_u = \gamma - y^\top a \leq 0$	ordinary pivot, u enters

Theorem: Right-hand side: shadow prices, range, dual simplex

$b' = b + \delta e_i$, $g = B^{-1}e_i$ (minus the tableau column of s_i). New constants $x_B + \delta g$. The basis stays optimal iff all ≥ 0:

$$\max_{k:\, g_k > 0} \Big(-\frac{x_{B(k)}}{g_k} \Big) \;\leq\; \delta \;\leq\; \min_{k:\, g_k < 0} \frac{x_{B(k)}}{-g_k},$$

and on that range $z^\star(\delta) = z^\star + y_i\delta$: y_i is the **shadow price**, the slope of the optimal value in b_i, valid inside the range only. The value function is concave piecewise linear in b.
Dual simplex step (outside the range): some constant $\beta_\ell < 0$ while all $r_j \leq 0$. If no coefficient in row ℓ is positive, the LP is infeasible. Otherwise choose the entering x_j with $q_{\ell j} > 0$ minimising $(-r_j)/q_{\ell j}$, solve row ℓ for x_j, substitute; the reduced costs stay ≤ 0 and z_0 does not increase. Repeat until all constants are ≥ 0.

Theorem: Objective change

Nonbasic $c_j \to c_j + t$: only $r_j \to r_j + t$; basis stays optimal iff $t \leq -r_j$ (the "break-even" increase is $-r_j$). Basic $c_{B(k)} \to c_{B(k)} + \varepsilon$: add $\varepsilon \times$(row of $x_{B(k)}$) to the z row; $z_0 \to z_0 + \varepsilon\beta_k$, $r_j \to r_j + \varepsilon q_{kj}$; basis stays optimal iff all new $r_j \leq 0$. The plan is unchanged inside the range; the value is not. The value function is convex piecewise linear in c.

Theorem: New activity

Column a, profit γ: $r_u = \gamma - y^\top a$ ($y^\top a$ is the resource cost of one unit at shadow prices). $r_u \leq 0$: do not make it, old plan optimal with $u = 0$. $r_u > 0$: tableau column $-B^{-1}a$, ratio test, pivot. B stays $m \times m$: a column is added, not a row.

15 Resource change: range, value, and beyond the range

Method

(a) Read $g = B^{-1}e_i$ = minus the column of s_i (unit vector if s_i is basic, then the range is one-sided and $y_i = 0$).

(b) One inequality per basic row: constant $+\delta g_k \geq 0$. Intersect. The range always contains 0.

(c) Inside: plan $x_B + \delta g$ (it moves!), value $z^\star + y_i\delta$.

(d) Outside: write the tableau with the new constants, find the negative one, dual simplex pivot, read the new plan, value and y.

Practice

Given. The optimal tableau of the running example: $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$. Resource 1 becomes $10 + \delta$.

Your task. (a) Range of δ and $z^\star(\delta)$. (b) Plan and value for $\delta = 1$. (c) $\delta = 4$: show the basis fails, reoptimise, give plan, value, y, and compare with the local formula. (d) Range and shadow price of resource 2 and of resource 3.

Solution

(a) $g = B^{-1}e_1 = -(-1, 1, -1)^\top = (1, -1, 1)^\top$. $4 + \delta \geq 0$, $2 - \delta \geq 0$, $3 + \delta \geq 0 \Rightarrow -3 \leq \delta \leq 2$. $z^\star = 22 + \delta$ ($y_1 = 1$).

(b) $\delta = 1$: $(x_1, x_2, s_3) = (5, 1, 4)$, $z = 23$. Same basis, different plan.

(c) $\delta = 4$: constants $(8, -2, 7)$: $x_2 < 0$, infeasible. Row $x_2 = -2 + s_1 - 2s_2$; positive coefficient only on s_1 (ratio $-r_{s_1}/1 = 1$): s_1 enters, x_2 leaves. $s_1 = 2 + x_2 + 2s_2$; substitute: $x_1 = 8 - s_1 + s_2 = 6 - x_2 - s_2$, $s_3 = 7 - s_1 + 2s_2 = 5 - x_2$, $z = 26 - s_1 - 2s_2 = 24 - x_2 - 4s_2$. New optimum $(6, 0)$, $z^\star = 24$, $y = (0, 4, 0)$: resource 1 is no longer binding ($s_1 = 2$), its price fell to 0. The local formula would say $22 + 4 = 26$: wrong outside the range.

(d) Resource 2: $g = -(1, -2, 2)^\top = (-1, 2, -2)^\top$: $4 - \delta \geq 0$, $2 + 2\delta \geq 0$, $3 - 2\delta \geq 0 \Rightarrow -1 \leq \delta \leq \frac{3}{2}$, $y_2 = 2$. Resource 3: s_3 basic, $g = e_3$: $3 + \delta \geq 0 \Rightarrow \delta \geq -3$, no upper limit, $y_3 = 0$ (adding market capacity is worthless, the market limit does not bind).

Gotcha

The row of a *basic slack* also limits the range (resource 2 above is cut by the s_3 row at $\frac{3}{2}$). The sign of g: minus the displayed column. "Same basis" never means "same plan": the plan is $B^{-1}b'$. Beyond the range the slope changes, often to 0.

16 Profit change

Practice

Given. Same tableau. (a) Profit of x_1 becomes $4+\varepsilon$. (b) Profit of x_2 becomes $3+\varepsilon$. (c) In the reading example of Part V, the profit of nonbasic x_2 becomes $2+t$.

Your task. Ranges, and what changes inside them.

Solution

(a) Add $\varepsilon \times (x_1$ row): $z = 22+4\varepsilon+(-1-\varepsilon)s_1+(-2+\varepsilon)s_2$. Conditions $-1-\varepsilon \le 0$, $-2+\varepsilon \le 0$: $-1 \le \varepsilon \le 2$, i.e. $c_1 \in [3,6]$. Plan $(4,2)$ unchanged, value $22+4\varepsilon$, $y = (1+\varepsilon, 2-\varepsilon, 0)$ moves.
(b) $z = 22+2\varepsilon+(-1+\varepsilon)s_1+(-2-2\varepsilon)s_2$: $\varepsilon \le 1$, $\varepsilon \ge -1$: $c_2 \in [2,4]$, value $22+2\varepsilon$.
(c) Nonbasic: $r_2 = 0 \to t$; stays optimal iff $t \le 0$. Any increase makes x_2 enter (it was already a tie).

Gotcha

A basic profit change moves y and *every* reduced cost: recheck all of them. For a basic variable the range often has no upper end (raising its profit never makes it leave) but here both ends are finite because two slacks compete. At an endpoint, two plans are optimal.

17 New activity

Practice

Given. Running example. A new product $u \geq 0$ uses 1 h of resource 1 and 2 h of resource 2, is not limited by the market, and earns γ.

Your task. (a) For which γ is the old plan still optimal? (b) For $\gamma = 6$ find the new optimum.

Solution

(a) $a = (1, 2, 0)^\top$, $y^\top a = 1 + 4 = 5$, $r_u = \gamma - 5$. Old plan optimal iff $\gamma \leq 5$.
(b) $\gamma = 6$: $r_u = 1$. $B^{-1}a = (1-2,\ -1+4,\ 1-4)^\top = (-1, 3, -3)^\top$, column $-B^{-1}a = (1, -3, 3)^\top$: $x_1 = 4 + u$, $x_2 = 2 - 3u$, $s_3 = 3 + 3u$. Only x_2 limits: $u = \frac{2}{3}$, x_2 leaves. $u = \frac{2}{3} + \frac{1}{3}s_1 - \frac{2}{3}s_2 - \frac{1}{3}x_2$; $x_1 = \frac{14}{3} - \frac{2}{3}s_1 + \frac{1}{3}s_2 - \frac{1}{3}x_2$; $s_3 = 5 - x_2$; $z = \frac{68}{3} - \frac{2}{3}s_1 - \frac{8}{3}s_2 - \frac{1}{3}x_2$. New optimum $(x_1, x_2, u) = (\frac{14}{3}, 0, \frac{2}{3})$, $z^\star = \frac{68}{3} \approx 22.67$, $y = (\frac{2}{3}, \frac{8}{3}, 0)$. Check: $2 \cdot \frac{14}{3} + \frac{2}{3} = 10$, $\frac{14}{3} + \frac{4}{3} = 6$.

Gotcha

$y^\top a$ is the cost of the resources the unit consumes, priced at shadow prices; a product is worth making iff its profit beats that. A new column needs no new row and no new slack.

18 Simultaneous changes

Practice

Given. Running example; resource 1 becomes $10 + \delta$ and the profit of x_1 becomes $4 + \varepsilon$ at the same time.

Your task. Range where the basis stays optimal, and $z^\star(\delta, \varepsilon)$. Is the total change the sum of the separate changes?

Solution

Feasibility depends on δ only: $-3 \leq \delta \leq 2$. Optimality depends on ε only: $-1 \leq \varepsilon \leq 2$. Basis optimal on the rectangle. Value $c_B(\varepsilon)^\top B^{-1} b(\delta) = (4 + \varepsilon)(4 + \delta) + 3(2 - \delta) = 22 + \delta + 4\varepsilon + \varepsilon\delta$. Not additive: the term $\varepsilon\delta$ appears because the extra profit applies to a changed quantity of x_1.

Part VIII

Theory: what to state and how to prove it

19 Proof skeletons you must be able to write

Method

- **Weak duality**: insert $y^\top Ax$; two inequalities, each using one feasibility and one sign.
- **Final row certifies optimality**: the identity $c^\top x = z_0 + \sum r_j x_j$ holds for every solution of $Ax = b$; with $x \geq 0$ and $r \leq 0$, $c^\top x \leq z_0$; the BFS attains z_0.
- **Uniqueness from strict signs**: $c^\top x = z_0$ forces every $r_j x_j = 0$, so $x_N = 0$, then $x_B = B^{-1}b$.
- **Ratio test**: $x_B(\theta) = x_B + \theta d_B$ coordinate by coordinate; only $d_{B(k)} < 0$ bounds θ; $Ad = 0$ keeps the equations; gain $c^\top d = r_j$ because $c^\top d = c_j + c_B^\top d_B = c_j - c_B^\top B^{-1} A_j$.
- **Shadow price**: b does not appear in r, so only feasibility can break; value $c_B^\top B^{-1}(b+\Delta b) = z^\star + y^\top \Delta b$.
- **y is dual optimal**: $r \leq 0$ on original columns gives $A^\top y \geq c$; on slack columns gives $y \geq 0$; $b^\top y = z^\star$; weak duality.
- **Complementary slackness**: $b^\top y - c^\top x = y^\top(b - Ax) + x^\top(A^\top y - c)$, a sum of nonnegative products; zero gap iff every product is zero; feasibility plus equal values is optimality.
- **Unbounded $\Rightarrow$ dual infeasible**: a dual-feasible y would give the upper bound $b^\top y$.
- **Termination**: strict increase without degeneracy, finitely many bases.
- **Converse of optimality needs nondegeneracy**: with $r_j > 0$ and $\theta_{\max} > 0$ there is a strictly better point; at a degenerate point $\theta_{\max}$ may be 0 and the converse fails.

Practice

Given. The tableau $x_1 = 4 - s_1 + s_2$, $x_2 = 2 + s_1 - 2s_2$, $s_3 = 3 - s_1 + 2s_2$, $z = 22 - s_1 - 2s_2$ of the running example.

Your task. (a) Prove from the tableau alone that 22 is the optimal value and that the optimum is unique. (b) Prove that $y = (1, 2, 0)$ is dual feasible using only the reduced costs. (c) State precisely what is wrong with "an optimal BFS always has all reduced costs ≤ 0".

Solution

(a) The z row holds for every solution of the equations. A feasible point has $s_1, s_2 \geq 0$, so $z = 22 - s_1 - 2s_2 \leq 22$; the point $(4, 2, 0, 0, 3)$ is feasible with $z = 22$. Uniqueness: $z = 22$ forces $s_1 = 2s_2 = 0$, hence $s_1 = s_2 = 0$, and the rows then give $x_1 = 4$, $x_2 = 2$, $s_3 = 3$: one point.
(b) $r_{s_1} = -y_1 = -1 \leq 0$ and $r_{s_2} = -y_2 = -2 \leq 0$ give $y_1, y_2 \geq 0$; s_3 basic gives $y_3 = 0 \geq 0$. For the original columns, $r_{x_1} = r_{x_2} = 0$ (basic) means $c_j = y^\top A_j$: $2y_1 + y_2 = 4$ and $y_1 + y_2 + y_3 = 3$ hold with equality, so $A^\top y \geq c$. Dual feasible; $b^\top y = 22$ matches, so it is dual optimal.
(c) It is true for a *nondegenerate* optimal BFS. At a degenerate optimum, a basis representing the point may have $r_j > 0$ with $\theta_{\max} = 0$ (the improving direction leaves the feasible set immediately); another basis of the same point passes the test. Example: $x + s = 0$, $\max x$: basis $\{s\}$ shows $r_x = 1 > 0$ at the optimal point 0.

Part IX

Quick checks: true or false, one line each

20 The bank

Practice

Your task. Decide, with the one-line reason.

1. A BFS has exactly m nonzero coordinates.
2. Every basis gives a BFS.
3. If a feasible basis has all nonbasic reduced costs ≤ 0, its solution is optimal.
4. If a BFS is optimal, every basis of it has reduced costs ≤ 0.
5. A positive reduced cost always allows a strictly better feasible point.
6. A zero nonbasic reduced cost at a nondegenerate optimum gives infinitely many optima.
7. A tableau with all reduced costs ≤ 0 is optimal.
8. An unbounded feasible region implies no finite optimum.
9. If the primal is unbounded, the dual is infeasible.
10. If the primal is infeasible, the dual is unbounded.
11. At an optimal pair, a tight primal row has a positive dual variable.
12. At an optimal pair, a positive primal variable has a tight dual row.
13. Changing b changes the reduced costs.
14. Changing c changes the optimal plan of the current basis.
15. The shadow price is the slope of $z^\star$ in b_i for every δ.
16. Inside the range of b_i, the optimal plan is unchanged.
17. The effect of two simultaneous changes is the sum of the separate effects.
18. Removing a redundant constraint can change the optimal value.
19. A new activity changes the old reduced costs.
20. The slack columns of the final tableau show B^{-1}.

Solution

1 F, at most m (degeneracy). 2 F, $B^{-1}b$ may have negatives. 3 T, the objective row bounds every feasible point. 4 F, only if nondegenerate. 5 F, $\theta_{\max}$ may be 0 (degenerate) or the direction may be unbounded (then infinitely better, still "better"): the false part is the degenerate case. 6 T, move along the direction up to $\theta_{\max} > 0$. 7 F, the constants must also be ≥ 0. 8 F, $\max -x_1$, $x_1 \geq 0$. 9 T, a feasible y would bound the primal. 10 F, the dual may also be infeasible. 11 F, both y_i and s_i may be 0. 12 T, $x_j t_j = 0$. 13 F, r does not involve b. 14 F, $B^{-1}b$ does not involve c; only the value and the reduced costs change. 15 F, only inside the range. 16 F, the plan is $B^{-1}b'$ and moves; the basis is unchanged. 17 F, there is a bilinear term. 18 F, the feasible set is unchanged. 19 F, it adds one reduced cost. 20 F, they show $-B^{-1}$ (equation-style tableau).